

DW_ram_r_w_a_lat

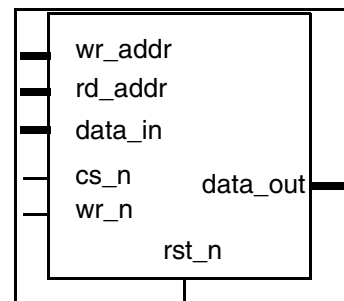
Asynchronous Dual-Port RAM (Latch-Based)

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Features and Benefits

- Parameterized word depth
- Parameterized data width
- Asynchronous static memory
- Parameterized reset implementation

Revision History



Description

DW_ram_r_w_a_lat implements a parameterized, asynchronous, dual-port static RAM.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
rst_n	1 bit	Input	Reset, active low
cs_n	1 bit	Input	Chip select, active low
wr_n	1 bit	Input	Write enable, active low
rd_addr	$\text{ceil}(\log_2[\text{depth}])$ bits	Input	Read address bus
wr_addr	$\text{ceil}(\log_2[\text{depth}])$ bits	Input	Write address bus
data_in	<i>data_width</i> bits	Input	Input data bus
data_out	<i>data_width</i> bits	Output	Output data bus

Table 1-2 Parameter Description

Parameter	Values	Description
data_width	1 to 256 Default: None	Width of <i>data_in</i> and <i>data_out</i> buses
depth	2 to 256 Default: None	Number of words in the memory array (address width)
rst_mode	0 or 1 Default: 1	Determines if the <i>rst_n</i> input is used. ■ 0 = <i>rst_n</i> initializes the RAM ■ 1 = <i>rst_n</i> is not connected

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW06.DW_RAM_R_W_A_LAT_CFG_SIM	VHDL simulation configuration
dw/dw06/src/DW_ram_r_w_a_lat_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_ram_r_w_a_lat.v	Verilog simulation model source code

The write data enters the RAM through the `data_in` input port and is read out at the `data_out` port. The RAM is constantly reading regardless of the state of `cs_n`.

The `rd_addr` and `wr_addr` ports are used to address the *depth* words in memory. For read addresses beyond the maximum depth (for example, `rd_addr` = 7 and *depth* = 6), the `data_out` bus is driven low. For `wr_addr` beyond the maximum depth, nothing occurs and the data is lost. No warnings are given during simulations when an address beyond the scope of *depth* is used.



Attention

This component contains enable signals for internal latches that are derived from the `wr_n` port. To keep hold times to a minimum, you should consider instances of this component to be individual floorplanning elements.

Chip Selection, Reading and Writing

The `cs_n` input is the chip select, active low signal, which enables data to be written to the RAM. The RAM is constantly reading, regardless of the state of `cs_n`.

When `cs_n` and `wr_n` (write enable, active low) are both low, the `data_in` is transparent to the internal memory cell being accessed (`data_in` = `data_out`). Therefore, during the period when `wr_n` and `cs_n` are low, a change in data in is reflected on the output of the internal memory cell being accessed.

If `rd_addr` and `wr_addr` are the same value and `wr_n` is low, data passes through the RAM (`data_in` = `data_out`). Data is captured into the memory cell on the low-to-high transition of `wr_n`.

When `cs_n` is high, writing to the RAM is disabled.

Reset

rst_n
This signal is an active-low input that initializes the RAM to zeros if the `rst_mode` parameter is set to 0, independent of the value of `cs_n`. If the `rst_mode` parameter is set to 1, `rst_n` does not affect the RAM and should be tied high or low. Synthesis optimizes the design, and does not use the `rst_n` signal.

 **Attention**

If the technology library being used does not contain an active low D-latch with clear, synthesis gates the inputs of a D-latch with the `rst_n` signal, increasing the area of the design.

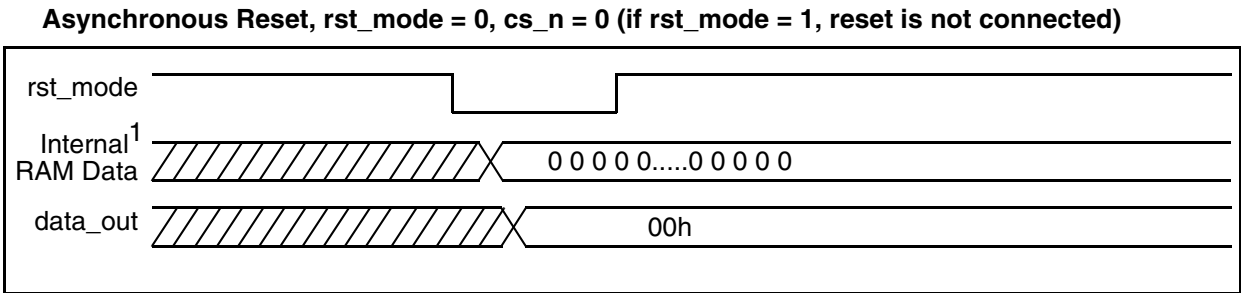
Application Notes

DW_ram_r_w_a_lat is intended to be used as a small scratch-pad memory or register file. Because DW_ram_r_w_a_lat is built from the cells within the ASIC cell library, it should be kept small to obtain an efficient implementation. If a larger memory is required, you should consider using a hard macro RAM from the ASIC library in use.

Timing Waveforms

The figures in this section show timing diagrams for various conditions of DW_ram_r_w_a_lat

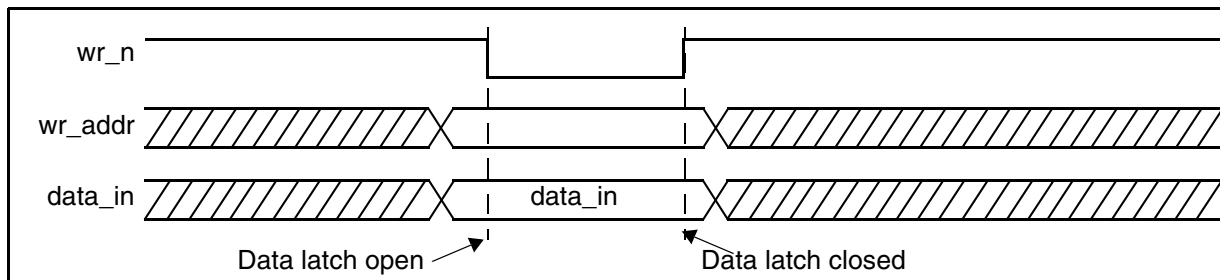
Figure 1-1 RAM Reset Timing Waveforms.



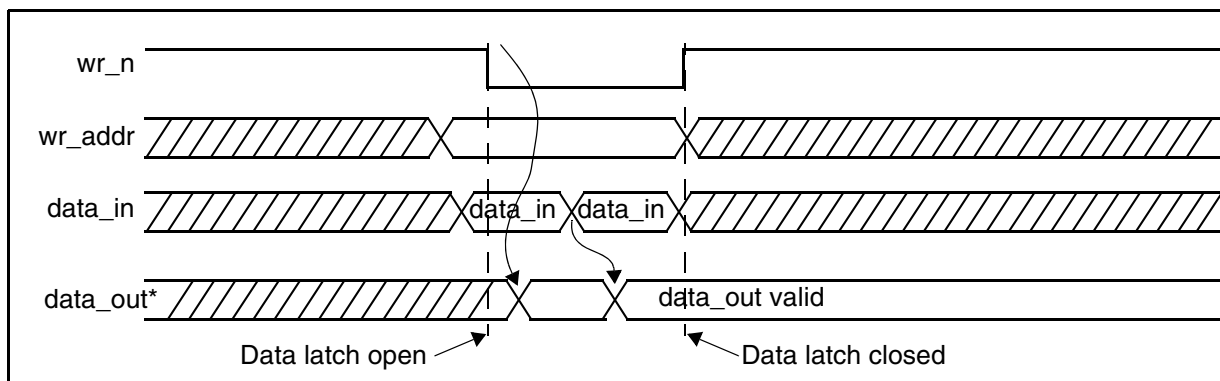
¹ Internal RAM Data is the array of memory bits; the memory is not available to users.

Figure 1-2 Instantiated RAM Timing Waveforms

Write Timing, cs_n = 0, Normal Data Input and Output

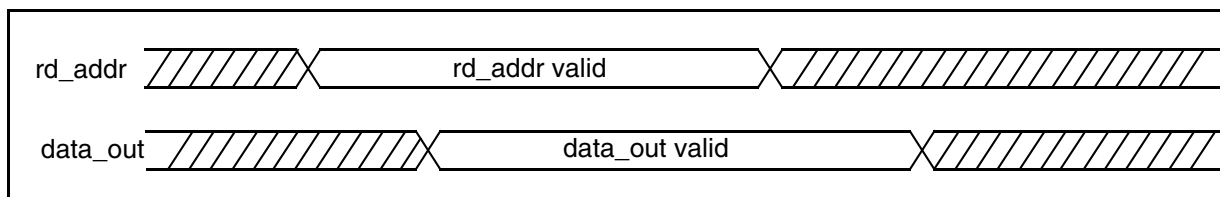


Write Timing, cs_n = 0, Changing Data Input and Output



* Write through is seen only when rd_addr = wr_addr.

Read Timing, address controlled, cs_n = don't care



Related Topics

- [Memory – Synchronous RAMs Listing](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW_ram_r_w_a_lat_inst is
  generic (inst_data_width : INTEGER := 8;
           inst_depth      : INTEGER := 8;
           inst_rst_mode   : INTEGER := 1 );
  port (inst_rst_n   : in std_logic;
        inst_cs_n    : in std_logic;
        inst_wr_n    : in std_logic;
        inst_rd_addr : in std_logic_vector(bit_width(inst_depth)-1 downto 0);
        inst_wr_addr : in std_logic_vector(bit_width(inst_depth)-1 downto 0);
        inst_data_in : in std_logic_vector(inst_data_width-1 downto 0);
        data_out_inst: out std_logic_vector(inst_data_width-1 downto 0) );
end DW_ram_r_w_a_lat_inst;

architecture inst of DW_ram_r_w_a_lat_inst is
begin

  -- Instance of DW_ram_r_w_a_lat
  U1 : DW_ram_r_w_a_lat
    generic map (data_width => inst_data_width,   depth => inst_depth,
                 rst_mode => inst_rst_mode )
    port map (rst_n => inst_rst_n,   cs_n => inst_cs_n,   wr_n => inst_wr_n,
              rd_addr => inst_rd_addr,   wr_addr => inst_wr_addr,
              data_in => inst_data_in,   data_out => data_out_inst );

end inst;

-- pragma translate_off
configuration DW_ram_r_w_a_lat_inst_cfg_inst of DW_ram_r_w_a_lat_inst is
  for inst
    end for; -- inst
end DW_ram_r_w_a_lat_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```
module DW_ram_r_w_a_lat_inst(inst_rst_n, inst_cs_n, inst_wr_n, inst_rd_addr,
                             inst_wr_addr, inst_data_in, data_out_inst );

    parameter data_width = 8;
    parameter depth = 8;
    parameter rst_mode = 1;
    `define bit_width_depth 3 // ceil(log2(depth))

    input inst_rst_n;
    input inst_cs_n;
    input inst_wr_n;
    input [`bit_width_depth-1 : 0] inst_rd_addr;
    input [`bit_width_depth-1 : 0] inst_wr_addr;
    input [data_width-1 : 0] inst_data_in;
    output [data_width-1 : 0] data_out_inst;

    // Instance of DW_ram_r_w_a_lat
    DW_ram_r_w_a_lat #(data_width, depth, rst_mode)
        U1 (.rst_n(inst_rst_n), .cs_n(inst_cs_n), .wr_n(inst_wr_n),
           .rd_addr(inst_rd_addr), .wr_addr(inst_wr_addr),
           .data_in(inst_data_in), .data_out(data_out_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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